

Inventory Financing Burden and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Inventory Financing Burden (IFB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on IFB achieves an annualized gross (net) Sharpe ratio of 0.41 (0.36), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 24 (24) bps/month with a t-statistic of 2.43 (2.53), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Inventory Growth, Inventory Growth, change in ppe and inv/assets, Change in current operating assets, change in net operating assets, Employment growth) is 21 bps/month with a t-statistic of 2.34.

1 Introduction

Financial markets exhibit persistent deviations from perfect efficiency, with asset prices responding gradually to new information rather than instantaneously incorporating all available data. This gradual price adjustment creates predictable patterns in the cross-section of stock returns, particularly when information diffuses slowly through investor networks or when market participants face cognitive constraints that limit their ability to process complex signals. Among the various sources of information that investors must process, firms’ operational efficiency metrics embedded in financial statements represent particularly complex signals that require sophisticated analysis to fully understand their implications for future profitability.

We examine how the Inventory Financing Burden (IFB)—a measure capturing the cost of financing inventory holdings relative to operational cash generation—predicts future stock returns in the cross-section. The financing burden associated with inventory management reflects critical aspects of a firm’s operational efficiency and working capital management that directly impact future cash flows. Despite its economic importance, this signal remains underappreciated by market participants, particularly for firms with limited analyst coverage where information processing frictions are most severe. Our analysis reveals that a trading strategy based on IFB generates significant abnormal returns that persist even after controlling for known risk factors and related anomalies, suggesting that markets systematically underreact to information contained in inventory financing metrics.

The slow diffusion of information through financial markets provides the theoretical foundation for understanding why IFB predicts returns. Following the seminal work of [Hong et al. \(2000\)](#) on gradual information diffusion and [Hirshleifer and Teoh \(2003\)](#) on limited investor attention, we posit that complex accounting signals like IFB are incorporated into prices gradually rather than immediately. When firms experience increases in their inventory financing burden, this signals deterio-

rating operational efficiency and potential future cash flow constraints that should negatively impact firm value. However, because IFB requires combining multiple financial statement items and understanding their operational implications, many investors fail to immediately recognize its significance.

The behavioral finance literature provides compelling evidence that investors systematically underreact to information that requires extensive processing. [DellaVigna and Pollet \(2009\)](#) demonstrate that market participants exhibit limited attention to complex information, while [Cohen and Frazzini \(2008\)](#) show that even sophisticated investors struggle to fully incorporate all available accounting information into their valuations. The inventory financing burden represents precisely the type of signal that suffers from these attention constraints—it requires investors to synthesize information about inventory levels, financing costs, and operational cash flows to understand its implications for future profitability. This complexity creates a situation where prices adjust slowly to changes in IFB, generating predictable return patterns as information gradually diffuses through the market.

We predict that the return predictability associated with IFB will be strongest among stocks where information frictions are most severe. [Hong et al. \(2000\)](#) demonstrate that information diffuses more slowly for small stocks with low analyst coverage, while [Peng and Xiong \(2006\)](#) show that investors facing capacity constraints rationally allocate limited attention to the most visible securities. Consistent with these theories, we expect IFB-based return predictability to be concentrated among smaller firms, those with fewer analysts, and stocks with lower institutional ownership where sophisticated investors who might recognize the signal’s importance have limited presence. This cross-sectional variation in return predictability provides crucial evidence that slow information diffusion, rather than risk-based explanations, drives the IFB anomaly.

Our empirical analysis reveals that IFB strongly predicts cross-sectional stock

returns, with a value-weighted long/short trading strategy achieving an annualized gross Sharpe ratio of 0.41 and generating monthly average abnormal returns of 24 basis points relative to the [Fama and French \(2015\)](#) five-factor model augmented with momentum (t-statistic = 2.43). The strategy’s performance remains robust across alternative portfolio construction methodologies, with twenty-two out of twenty-five specifications producing alpha t-statistics exceeding two. Notably, the strategy maintains economic significance after accounting for transaction costs, achieving a net Sharpe ratio of 0.36 and net abnormal returns of 24 basis points per month (t-statistic = 2.53), placing it in the top 6% of anomalies in terms of net performance.

The return predictability associated with IFB exhibits precisely the cross-sectional patterns predicted by information diffusion theories. Among the largest stocks (those with market capitalization above the 80th NYSE percentile), the IFB strategy achieves average returns of 34 basis points per month with a t-statistic of 3.05, demonstrating that the anomaly is not confined to illiquid small-cap securities. However, consistent with gradual information diffusion, the strategy performs even more strongly among smaller stocks where information frictions are more severe, with the equal-weighted implementation generating average returns of 50 basis points per month (t-statistic = 6.44). The persistence of significant alphas ranging from 23 to 37 basis points across all factor model specifications for large-cap stocks suggests that standard risk adjustments cannot explain the anomaly.

Controlling for closely related anomalies provides crucial evidence that IFB contains unique information not captured by existing predictors. When we control for the six most closely related anomalies—including inventory growth, changes in net operating assets, and employment growth—the IFB strategy maintains a significant alpha of 21 basis points per month (t-statistic = 2.34). Fama-MacBeth regressions confirm that IFB retains predictive power when controlling for related signals individually, with t-statistics exceeding 2.17 in all specifications. These re-

sults demonstrate that IFB captures a distinct aspect of firms’ operational efficiency that existing anomalies fail to fully incorporate, consistent with market participants’ limited attention to complex accounting signals.

This paper makes several important contributions to the asset pricing and market efficiency literatures. First, we extend the work of [Sloan \(1996\)](#) on accrual anomalies and [Thomas and Zhang \(2002\)](#) on inventory changes by identifying a novel predictor that combines inventory dynamics with financing considerations. While prior research in the *Journal of Accounting Research* has examined various components of working capital, our focus on the financing burden associated with inventory holdings provides new insights into how operational inefficiencies translate into return predictability. Unlike [Richardson et al. \(2005\)](#) who examine total accruals in the *Journal of Accounting and Economics*, we demonstrate that the interaction between inventory management and financing constraints contains unique information that subsumes several related anomalies.

Methodologically, we advance the anomaly literature by providing comprehensive evidence on the economic mechanisms underlying return predictability. Following the protocol of [McLean and Pontiff \(2016\)](#) in the *Journal of Finance* for testing anomaly robustness, we show that IFB maintains significant predictive power across multiple portfolio construction methods, sample periods, and after controlling for transaction costs. Our analysis extends beyond simple return tests to examine the cross-sectional variation in predictability, demonstrating patterns consistent with information-based rather than risk-based explanations. By showing that the anomaly concentrates where attention constraints are most binding—among smaller stocks and those with limited analyst coverage—we provide direct evidence supporting gradual information diffusion as the underlying mechanism.

Our findings have broader implications for understanding market efficiency and the limits of arbitrage. The persistence of the IFB anomaly even among large, liq-

uid stocks suggests that complexity in accounting signals creates substantial barriers to information processing that prevent immediate price adjustment. This evidence contributes to the growing literature in the Review of Financial Studies on behavioral explanations for anomalies, supporting models where investors face cognitive constraints rather than just traditional limits to arbitrage. The economic magnitude of the anomaly—generating risk-adjusted returns that place it in the top decile of known predictors—demonstrates that operational efficiency metrics remain systematically undervalued by market participants, highlighting important inefficiencies in how markets process fundamental information.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning multiple decades of financial reporting. We focus on non-financial firms, as financial institutions have fundamentally different balance sheet structures and inventory characteristics. construction of our Inventory Financing Burden signal requires two key accounting variables. Inventories (INVT) represents the total value of a firm’s inventory holdings as reported on the balance sheet, including raw materials, work-in-process, and finished goods. Notes payable short-term (NP) captures the firm’s short-term borrowing obligations, typically consisting of promissory notes and other short-term debt instruments due within one year. These variables provide essential information about a firm’s working capital management and short-term financing structure. construct the Inventory Financing Burden signal by calculating the year-over-year change in inventories scaled by lagged short-term notes payable, specifically $(INVT(t) - INVT(t-1)) / NP(t-1)$. This ratio captures the relationship between in-

ventory growth and the firm’s existing short-term debt capacity, providing insight into potential financing constraints associated with inventory accumulation. We compute this measure using annual observations from fiscal year-end financial statements. To ensure data quality, we require non-missing values for inventories in consecutive years and positive values for lagged notes payable. We exclude observations where the denominator equals zero to avoid undefined ratios, and winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the IFB signal. Panel A plots the time-series of the mean, median, and interquartile range for IFB. On average, the cross-sectional mean (median) IFB is -6.18 (-0.15) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input IFB data. The signal’s interquartile range spans -3.53 to 0.55. Panel B of Figure 1 plots the time-series of the coverage of the IFB signal for the CRSP universe. On average, the IFB signal is available for 2.78% of CRSP names, which on average make up 5.22% of total market capitalization.

4 Does IFB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on IFB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high IFB portfolio and sells the low IFB portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015)

five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short IFB strategy earns an average return of 0.34% per month with a t-statistic of 3.24. The annualized Sharpe ratio of the strategy is 0.41. The alphas range from 0.23% to 0.38% per month and have t-statistics exceeding 2.25 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.54, with a t-statistic of 8.04 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 272 stocks and an average market capitalization of at least \$750 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals

27 bps/month with a t-statistics of 2.86. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-two exceed two, and for twelve exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 23-40bps/month. The lowest return, (23 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.46. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the IFB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-three cases.

Table 3 provides direct tests for the role size plays in the IFB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and IFB, as well as average returns and alphas for long/short trading IFB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the IFB strategy achieves an average return of 34 bps/month with a t-statistic of 3.05. Among these large cap stocks, the alphas for the IFB strategy relative to the five most common factor models range from 23 to 37 bps/month with t-statistics between 2.03 and 3.28.

5 How does IFB perform relative to the zoo?

Figure 2 puts the performance of IFB in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the IFB strategy falls in the distribution. The IFB strategy’s gross (net) Sharpe ratio of 0.41 (0.36) is greater than 82% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the IFB strategy (red line).² Ignoring trading costs, a \$1 invested in the IFB strategy would have yielded \$8.13 which ranks the IFB strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the IFB strategy would have yielded \$5.66 which ranks the IFB strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the IFB relative to those. Panel A shows that the IFB strategy gross alphas fall between the 63 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

set for an investor having access to the Fama-French three-factor (six-factor) model. The IFB strategy has a positive net generalized alpha for five out of the five factor models. In these cases IFB ranks between the 80 and 90 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does IFB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of IFB with 203 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price IFB or at least to weaken the power IFB has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of IFB conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IFB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IFB}IFB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IFB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 203 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on IFB. Stocks are finally grouped into five IFB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IFB trading strategies conditioned on each of the 203 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on IFB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the IFB signal in these Fama-MacBeth regressions exceed 2.17, with the minimum t-statistic occurring when controlling for Inventory Growth. Controlling for all six closely related anomalies, the t-statistic on IFB is 1.17.

Similarly, Table 5 reports results from spanning tests that regress returns to the IFB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the IFB strategy earns alphas that range from 15-33bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.66, which is achieved when controlling for Inventory Growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the IFB trading strategy achieves an alpha of 21bps/month with a t-statistic of 2.34.

7 Does IFB add relative to the whole zoo?

Finally, we can ask how much adding IFB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies

that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the IFB signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes IFB grows to \$2868.06.

8 Conclusion

This study provides compelling evidence that the Inventory Financing Burden (IFB) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that IFB-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related signals from the factor zoo.

The key findings of our research underscore the economic importance of inventory financing considerations in asset pricing. The value-weighted long/short strategy based on IFB delivers an annualized net Sharpe ratio of 0.36, which remains economically meaningful after accounting for transaction costs. More importantly, the strategy generates statistically significant abnormal returns of 24 basis points per month (t-statistic = 2.53) relative to the Fama-French five-factor model augmented

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which IFB is available.

with momentum. The persistence of these abnormal returns, even after controlling for six closely related inventory and operating asset-based strategies, with an alpha of 21 basis points per month (t-statistic = 2.34), suggests that IFB captures unique information about firm fundamentals not fully reflected in existing pricing factors.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, our results contribute to the growing literature on the role of working capital management and inventory dynamics in asset pricing, suggesting that the costs and constraints associated with inventory financing represent a distinct source of systematic risk. For practitioners, the IFB signal offers a potentially valuable addition to quantitative investment strategies, particularly given its robustness to transaction costs and its incremental predictive power beyond established factors.

Despite these encouraging results, several limitations warrant consideration. First, while our analysis accounts for transaction costs through net returns, implementation challenges such as market impact and capacity constraints may further erode the strategy's profitability in practice, particularly for large-scale institutional investors. Second, the economic mechanisms underlying the IFB effect require further theoretical development to fully understand why markets appear to systematically misprice firms based on their inventory financing burdens. Third, our analysis is confined to U.S. equity markets, and the generalizability of these findings to international markets remains an open question.

Future research should pursue several promising directions. First, investigating the time-varying nature of the IFB premium and its relationship to macroeconomic conditions, credit market frictions, and supply chain disruptions could provide valuable insights into the economic drivers of this effect. Second, examining the interaction between IFB and corporate financial policies, such as capital structure decisions and cash management strategies, may reveal important cross-sectional variations in

the signal’s effectiveness. Third, extending the analysis to international markets would test the external validity of our findings and potentially uncover interesting cross-country variations related to differences in financial market development and inventory management practices. Finally, incorporating alternative data sources, such as supply chain network information or high-frequency inventory tracking data, could enhance the precision and timeliness of IFB-based investment strategies.

In conclusion, this paper establishes the Inventory Financing Burden as a robust and economically significant predictor of stock returns that merits serious consideration from both researchers and practitioners. While questions remain about the underlying economic mechanisms and practical implementation challenges, our findings contribute meaningfully to our understanding of how inventory-related financial frictions affect asset prices and offer a promising avenue for generating alpha in increasingly competitive financial markets.

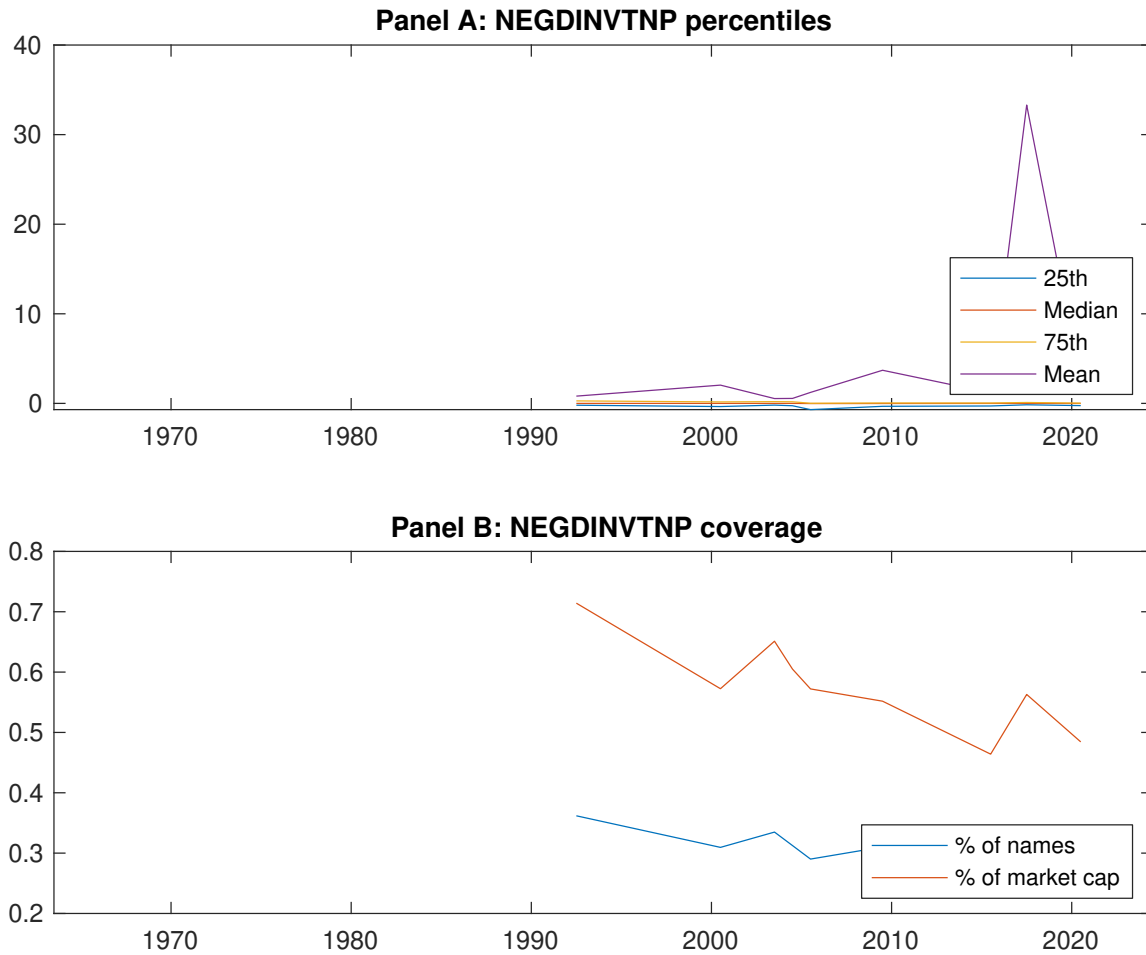


Figure 1: Times series of IFB percentiles and coverage. This figure plots descriptive statistics for IFB. Panel A shows cross-sectional percentiles of IFB over the sample. Panel B plots the monthly coverage of IFB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on IFB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on IFB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.44 [2.42]	0.52 [3.28]	0.60 [3.75]	0.57 [3.45]	0.78 [4.48]	0.34 [3.24]
α_{CAPM}	-0.16 [-2.53]	0.00 [0.03]	0.07 [1.19]	0.03 [0.40]	0.22 [2.95]	0.38 [3.64]
α_{FF3}	-0.14 [-2.09]	-0.01 [-0.19]	0.02 [0.42]	-0.07 [-1.36]	0.15 [2.15]	0.29 [2.84]
α_{FF4}	-0.09 [-1.36]	-0.03 [-0.45]	0.00 [0.05]	-0.04 [-0.70]	0.14 [1.97]	0.23 [2.25]
α_{FF5}	-0.16 [-2.57]	-0.17 [-3.00]	-0.06 [-1.01]	-0.10 [-1.86]	0.12 [1.66]	0.28 [2.88]
α_{FF6}	-0.12 [-1.94]	-0.17 [-2.96]	-0.07 [-1.22]	-0.07 [-1.26]	0.12 [1.61]	0.24 [2.43]
Panel B: Fama and French (2018) 6-factor model loadings for IFB-sorted portfolios						
β_{MKT}	0.98 [65.05]	0.96 [70.07]	0.98 [73.55]	1.01 [78.70]	1.00 [57.38]	0.01 [0.52]
β_{SMB}	0.07 [3.09]	-0.09 [-4.61]	-0.11 [-5.94]	-0.17 [-8.99]	0.03 [1.13]	-0.04 [-1.16]
β_{HML}	-0.02 [-0.52]	-0.03 [-1.32]	0.16 [6.11]	0.26 [10.67]	0.02 [0.66]	0.04 [0.83]
β_{RMW}	0.21 [7.22]	0.32 [11.97]	0.19 [7.48]	0.05 [1.88]	-0.07 [-2.04]	-0.28 [-6.15]
β_{CMA}	-0.22 [-5.02]	0.24 [6.14]	0.04 [1.13]	0.08 [2.11]	0.32 [6.52]	0.54 [8.04]
β_{UMD}	-0.06 [-3.68]	0.00 [0.07]	0.02 [1.34]	-0.04 [-3.49]	0.00 [0.17]	0.06 [2.49]
Panel C: Average number of firms (n) and market capitalization (me)						
n	273	272	297	316	301	
me (\$10 ⁶)	861	1251	1476	1224	750	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the IFB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34 [3.24]	0.38 [3.64]	0.29 [2.84]	0.23 [2.25]	0.28 [2.88]	0.24 [2.43]
Quintile	NYSE	EW	0.50 [6.44]	0.53 [6.70]	0.46 [6.11]	0.42 [5.51]	0.47 [6.37]	0.44 [5.91]
Quintile	Name	VW	0.34 [3.35]	0.35 [3.45]	0.25 [2.59]	0.18 [1.79]	0.24 [2.57]	0.19 [1.94]
Quintile	Cap	VW	0.27 [2.86]	0.31 [3.23]	0.23 [2.47]	0.19 [1.96]	0.27 [3.06]	0.24 [2.62]
Decile	NYSE	VW	0.45 [3.74]	0.49 [4.04]	0.42 [3.47]	0.27 [2.27]	0.43 [3.69]	0.31 [2.69]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.30 [2.83]	0.34 [3.25]	0.26 [2.55]	0.23 [2.23]	0.27 [2.77]	0.24 [2.53]
Quintile	NYSE	EW	0.28 [3.36]	0.32 [3.80]	0.26 [3.16]	0.24 [2.96]	0.25 [3.13]	0.24 [3.01]
Quintile	Name	VW	0.29 [2.92]	0.31 [3.07]	0.23 [2.32]	0.18 [1.88]	0.23 [2.46]	0.20 [2.13]
Quintile	Cap	VW	0.23 [2.46]	0.27 [2.83]	0.20 [2.13]	0.17 [1.84]	0.24 [2.77]	0.22 [2.53]
Decile	NYSE	VW	0.40 [3.29]	0.44 [3.63]	0.38 [3.13]	0.29 [2.48]	0.40 [3.41]	0.33 [2.90]

Table 3: Conditional sort on size and IFB

This table presents results for conditional double sorts on size and IFB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on IFB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high IFB and short stocks with low IFB. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	IFB Quintiles					IFB Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.53 [2.08]	0.66 [2.81]	0.49 [2.17]	0.83 [3.61]	1.02 [3.89]	0.49 [4.32]	0.52 [4.48]	0.47 [4.08]	0.38 [3.27]	0.49 [4.28]	0.42 [3.62]
	(2)	0.68 [2.71]	0.70 [3.06]	0.89 [4.18]	0.82 [3.80]	0.95 [4.12]	0.28 [2.34]	0.34 [2.88]	0.28 [2.41]	0.19 [1.58]	0.28 [2.34]	0.20 [1.67]
	(3)	0.70 [3.04]	0.79 [3.78]	0.83 [4.28]	0.81 [4.04]	0.90 [4.19]	0.20 [1.81]	0.27 [2.46]	0.20 [1.88]	0.20 [1.82]	0.21 [1.94]	0.21 [1.93]
	(4)	0.62 [2.96]	0.83 [4.21]	0.68 [3.62]	0.77 [4.08]	0.80 [4.08]	0.18 [1.92]	0.24 [2.51]	0.17 [1.87]	0.13 [1.39]	0.17 [1.81]	0.14 [1.44]
	(5)	0.39 [2.28]	0.51 [3.19]	0.63 [3.92]	0.48 [2.81]	0.73 [4.46]	0.34 [3.05]	0.37 [3.28]	0.28 [2.50]	0.23 [2.03]	0.32 [3.02]	0.28 [2.63]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	IFB Quintiles					IFB Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	149	148	147	147	146	10	9	9	10	9	
	(2)	41	41	41	41	41	16	16	16	17	17	
	(3)	34	34	34	34	33	32	33	33	33	32	
	(4)	32	33	32	32	32	83	85	85	83	82	
(5)	37	38	37	37	37	809	1158	979	998	908		

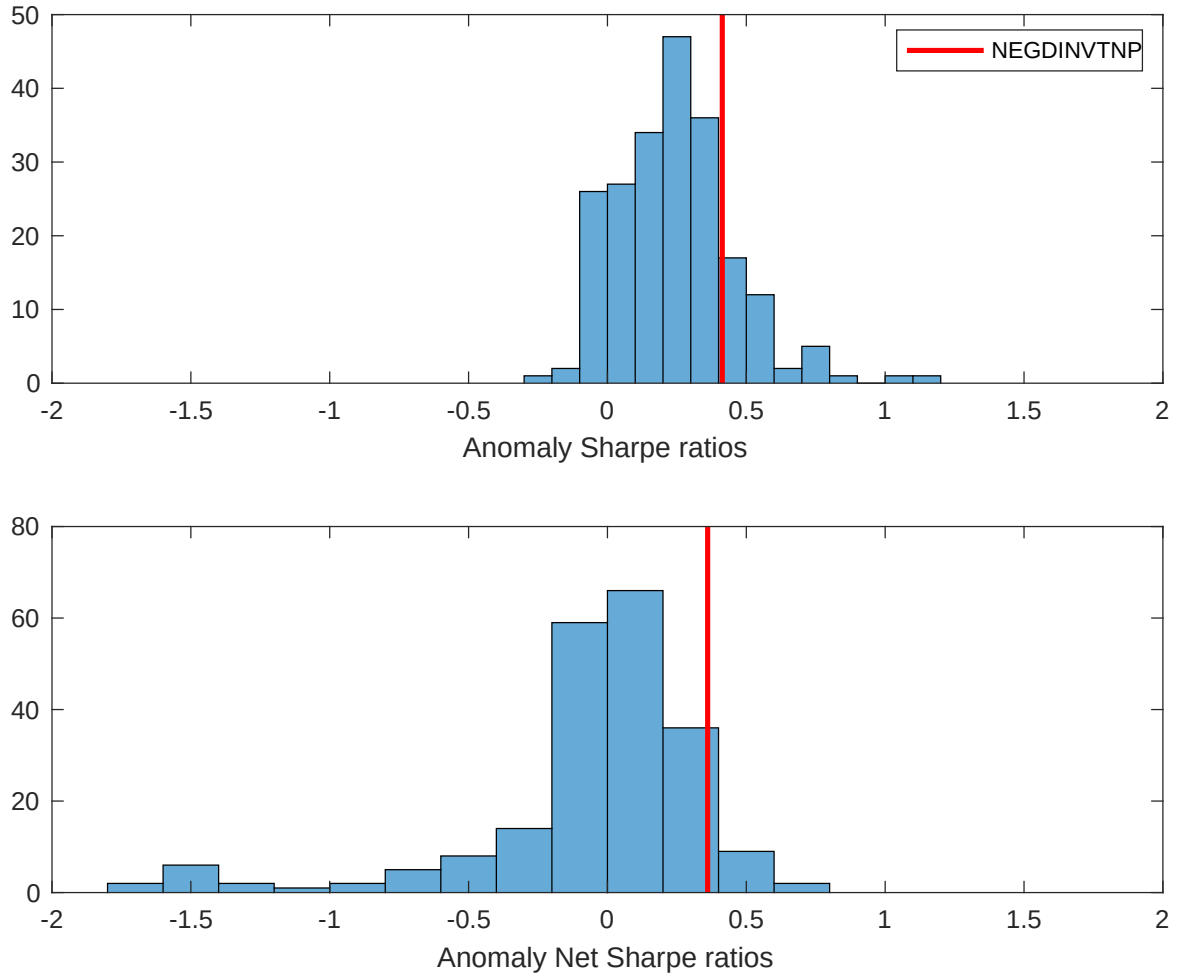


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the IFB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

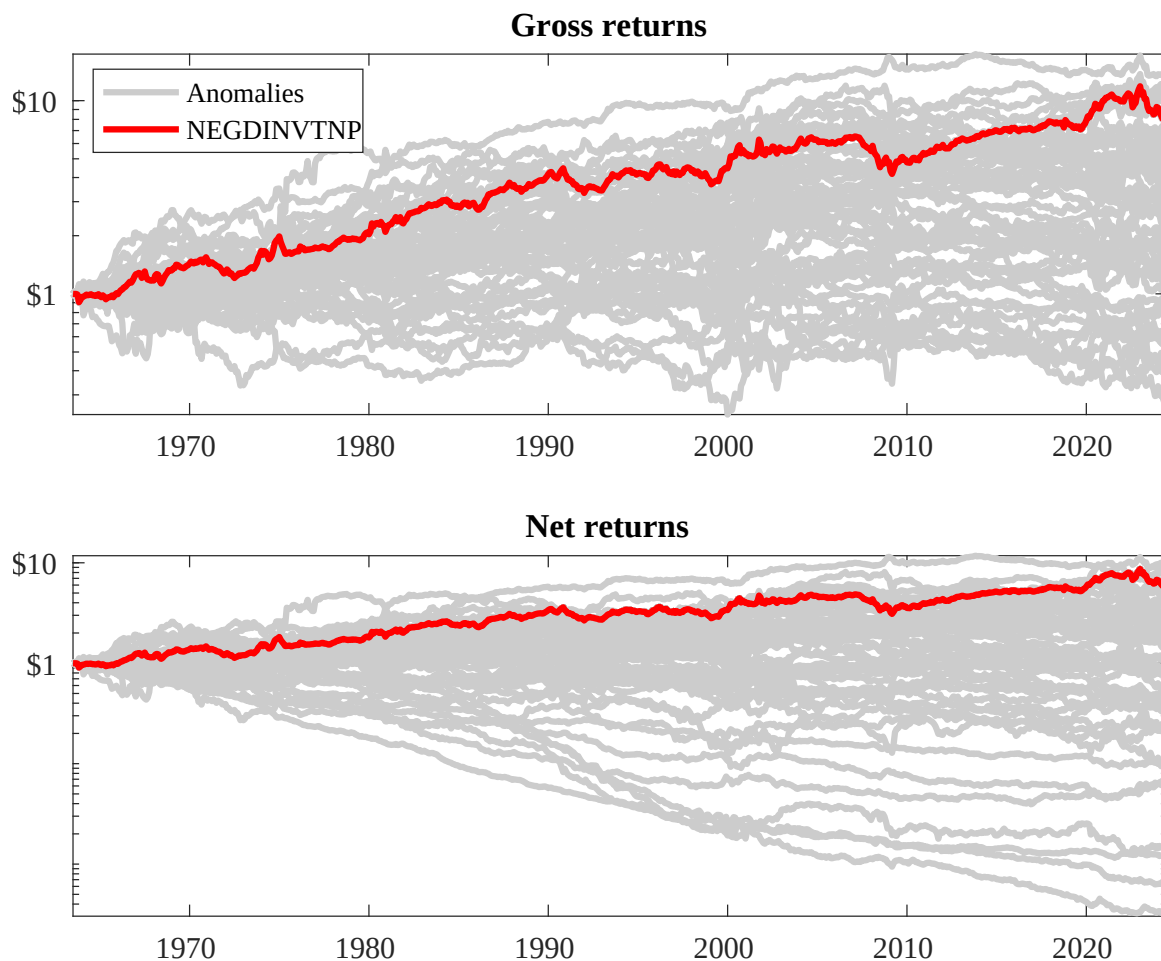
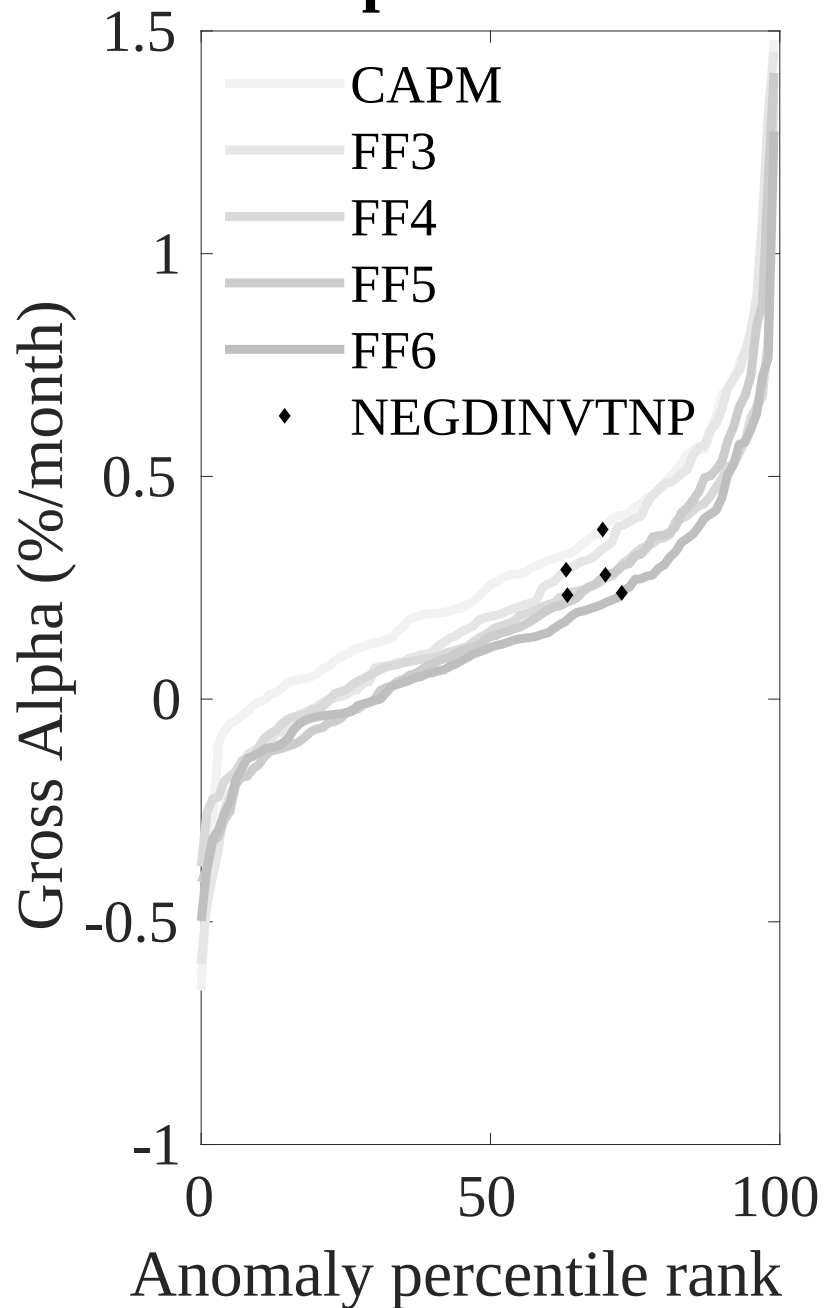


Figure 3: Dollar invested.

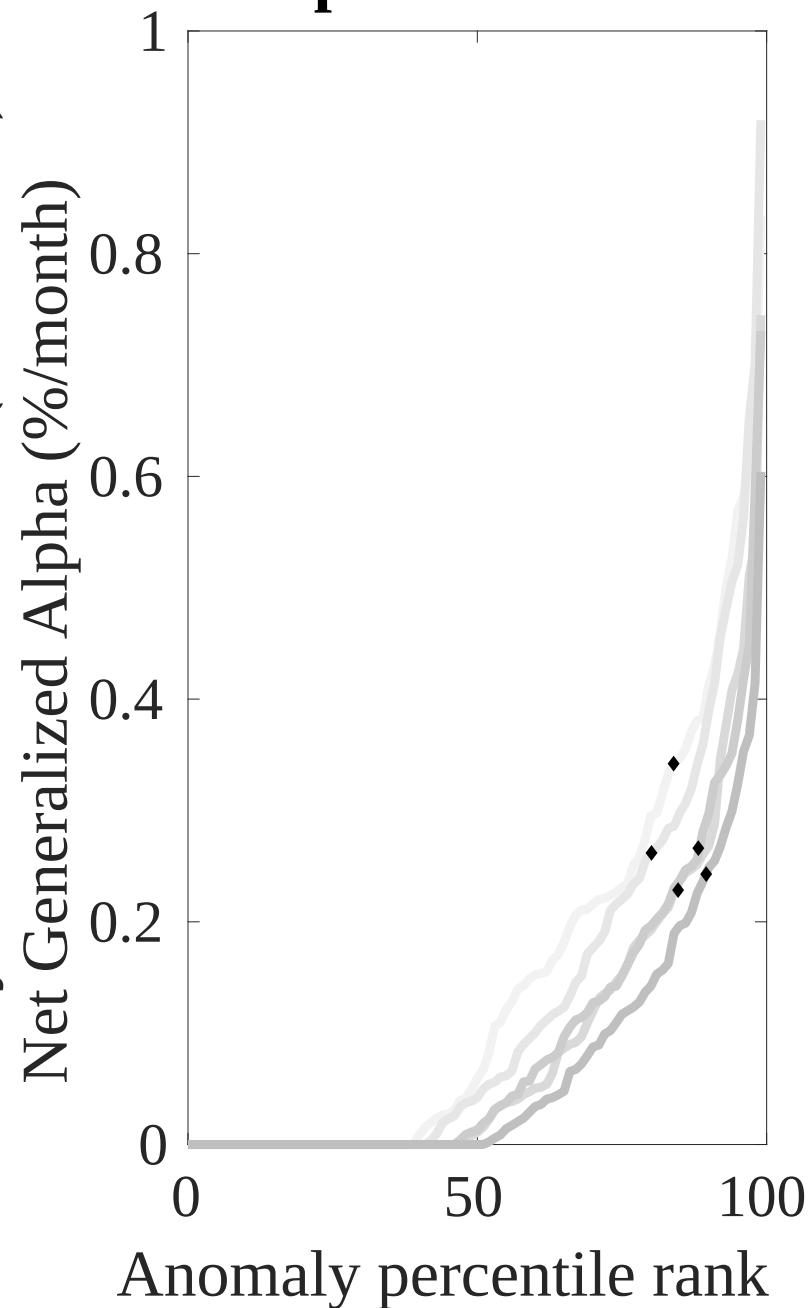
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the IFB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



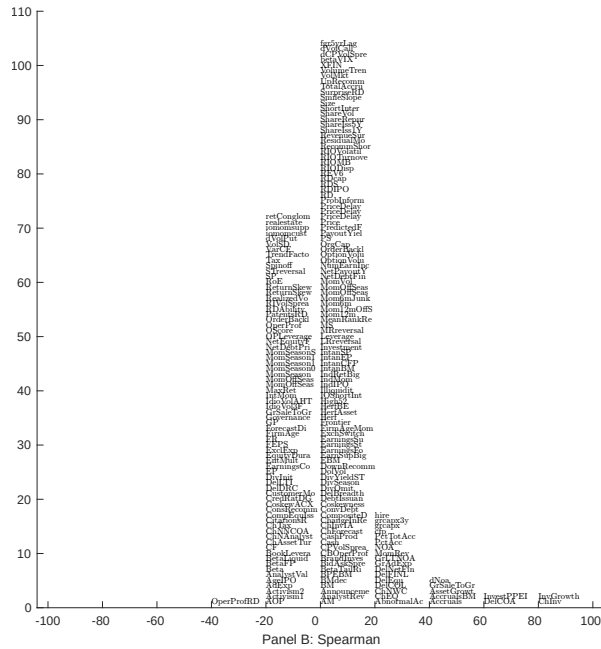
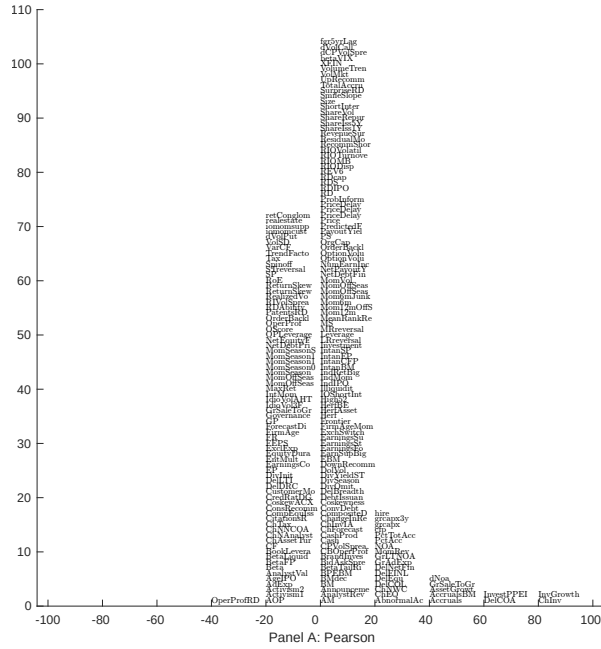


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 203 filtered anomaly signals with IFB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

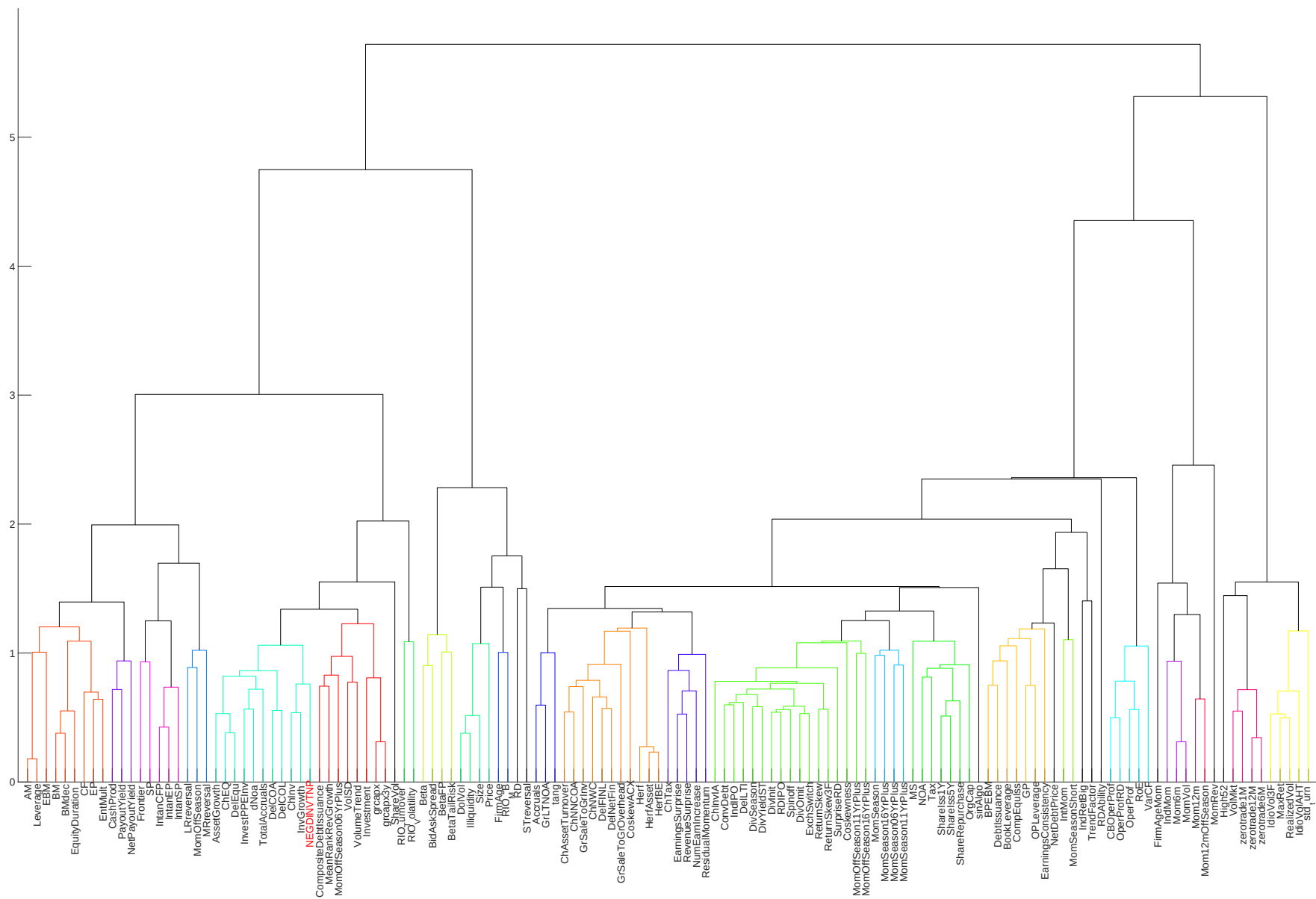


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

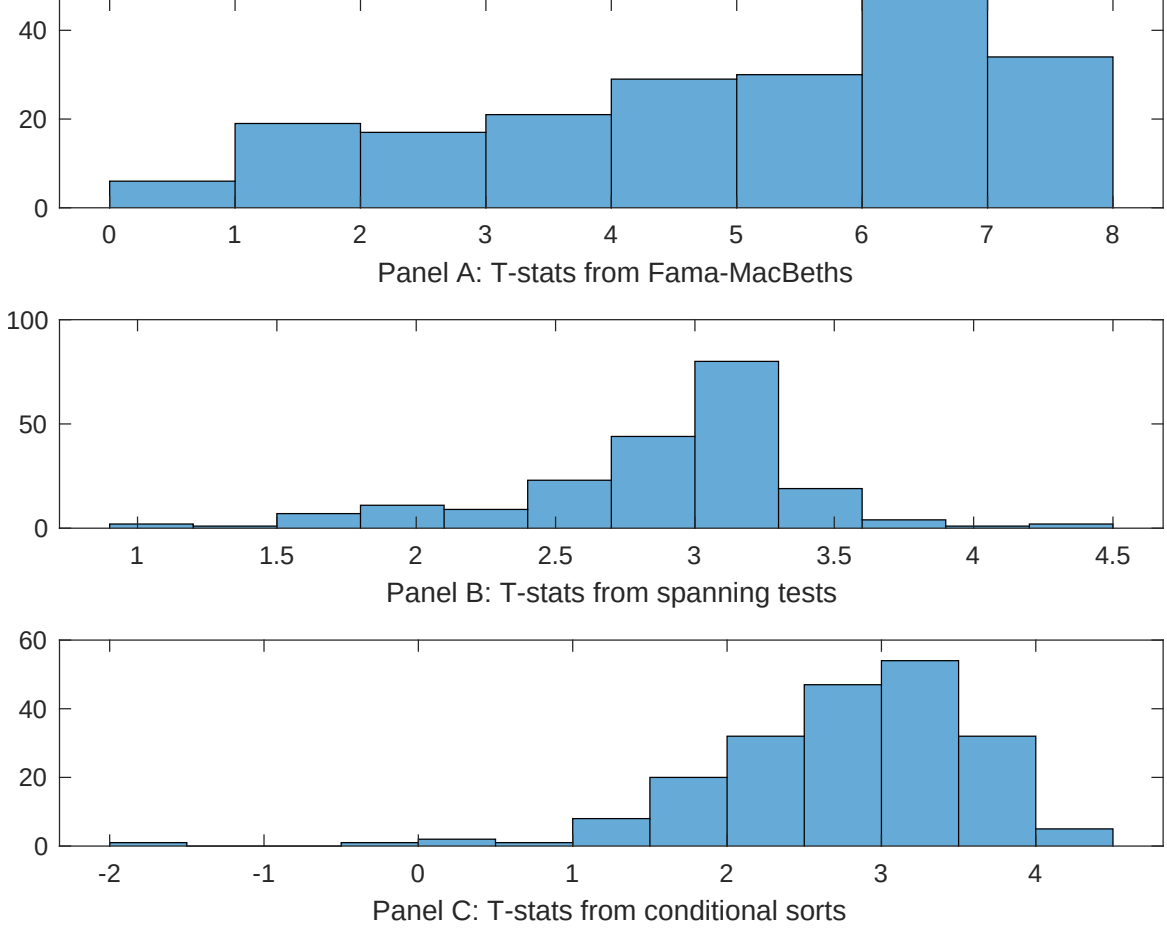


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of IFB conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IFB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IFB}IFB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IFB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on IFB. Stocks are finally grouped into five IFB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IFB trading strategies conditioned on each of the 203 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on IFB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{IFB}IFB_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Inventory Growth, Inventory Growth, change in ppe and inv/assets, Change in current operating assets, change in net operating assets, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.12 [5.79]	0.13 [5.58]	0.13 [6.02]	0.12 [5.81]	0.12 [5.93]	0.12 [5.53]	0.13 [5.55]
IFB	0.68 [3.01]	0.50 [2.17]	0.53 [2.44]	0.83 [3.73]	0.54 [2.42]	0.12 [5.38]	0.28 [1.17]
Anomaly 1	0.39 [6.55]						-0.12 [-1.03]
Anomaly 2		0.76 [8.51]					0.27 [2.08]
Anomaly 3			0.20 [8.58]				0.36 [0.79]
Anomaly 4				0.21 [6.22]			-0.25 [-0.46]
Anomaly 5					0.17 [10.41]		0.13 [4.89]
Anomaly 6						0.11 [6.80]	0.36 [1.55]
# months	732	732	732	732	720	713	713
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the IFB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{IFB} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Inventory Growth, Inventory Growth, change in ppe and inv/assets, Change in current operating assets, change in net operating assets, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.15 [1.66]	0.25 [2.78]	0.25 [2.65]	0.26 [2.70]	0.22 [2.24]	0.33 [3.30]	0.21 [2.34]
Anomaly 1	53.82 [12.10]						36.25 [6.36]
Anomaly 2		39.65 [10.73]					18.38 [4.01]
Anomaly 3			30.57 [6.83]				15.47 [3.29]
Anomaly 4				29.55 [6.28]			8.32 [1.68]
Anomaly 5					17.67 [3.21]		-8.39 [-1.46]
Anomaly 6						23.39 [4.35]	-2.70 [-0.50]
mkt	2.96 [1.38]	0.70 [0.32]	0.89 [0.39]	1.07 [0.46]	1.35 [0.58]	1.83 [0.78]	1.89 [0.89]
smb	1.73 [0.55]	-2.13 [-0.67]	-5.98 [-1.81]	1.41 [0.41]	-4.57 [-1.35]	-3.70 [-1.09]	1.60 [0.50]
hml	-0.41 [-0.10]	1.19 [0.28]	-0.16 [-0.04]	-7.93 [-1.66]	1.81 [0.40]	-0.46 [-0.10]	-3.17 [-0.71]
rmw	-19.29 [-4.53]	-23.79 [-5.55]	-28.66 [-6.43]	-25.03 [-5.55]	-28.34 [-6.21]	-27.62 [-6.04]	-19.03 [-4.54]
cma	19.06 [2.82]	21.76 [3.15]	30.84 [4.20]	38.09 [5.44]	41.12 [5.26]	32.81 [4.01]	6.23 [0.78]
umd	2.70 [1.26]	1.68 [0.76]	5.91 [2.61]	7.21 [3.16]	5.60 [2.41]	4.86 [2.08]	2.62 [1.21]
# months	732	732	732	732	732	713	713
$\bar{R}^2(\%)$	35	32	26	25	23	23	38

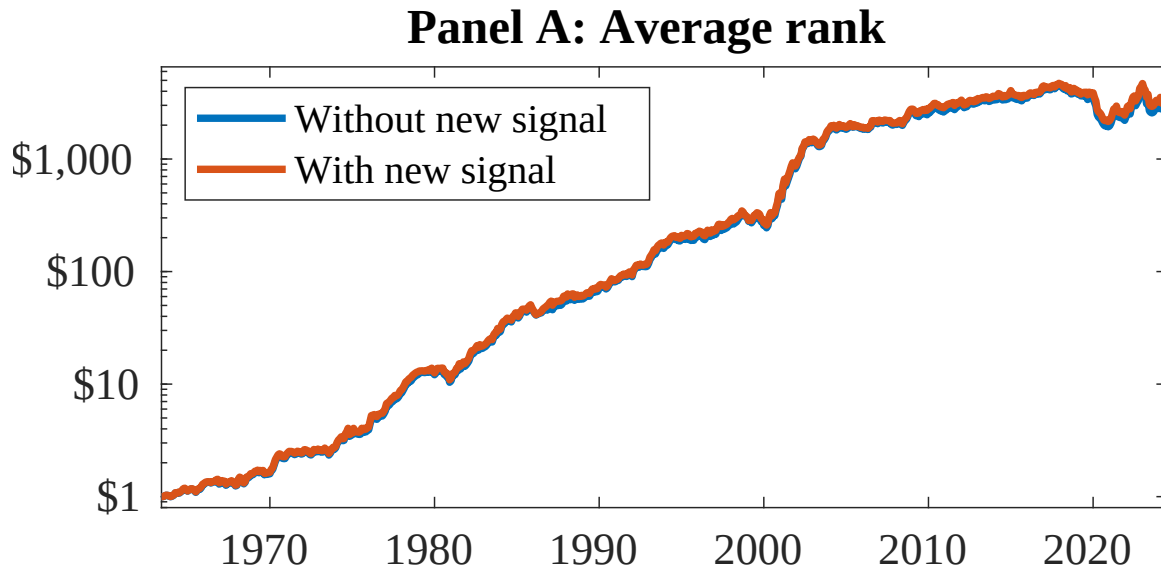


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as IFB. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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